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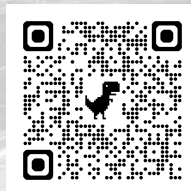
# Subspace Alignment for Vision-Language Model Test-time Adaptation

Zhichen Zeng<sup>1</sup>, Wenxuan Bao<sup>1</sup>, Xiao Lin<sup>1</sup>, Ruizhong Qiu<sup>1</sup>, Tianxin Wei<sup>1</sup>, Xuying Ning<sup>1</sup>,  
Yuchen Yan<sup>2</sup>, Chen Luo<sup>2</sup>, Monica Xiao Cheng<sup>2</sup>, Jingrui He<sup>1</sup>, Hanghang Tong<sup>1</sup>

<sup>1</sup> University of Illinois Urbana-Champaign    <sup>2</sup> Amazon

Email: [zhichenz@illinois.edu](mailto:zhichenz@illinois.edu)

Paper



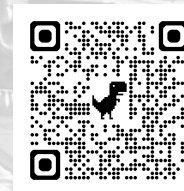
Code



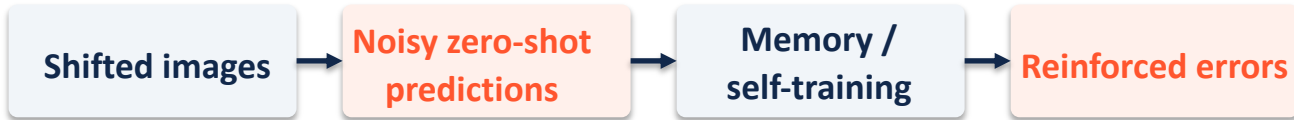
Slides



Poster

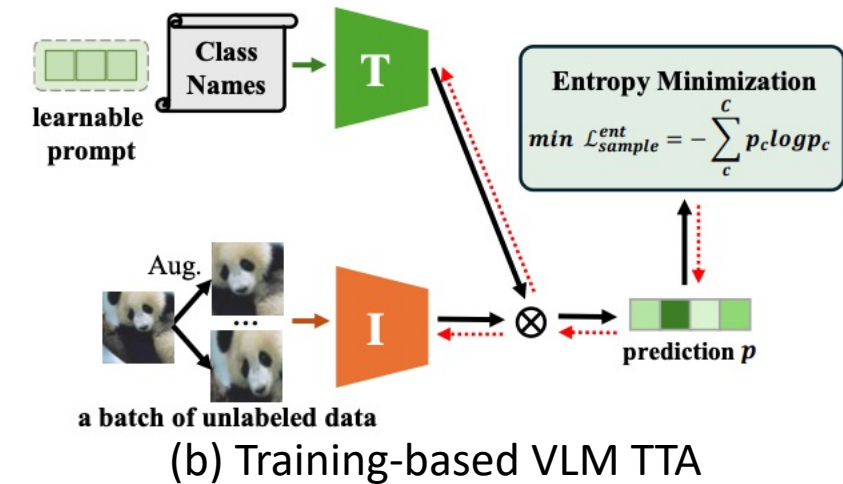
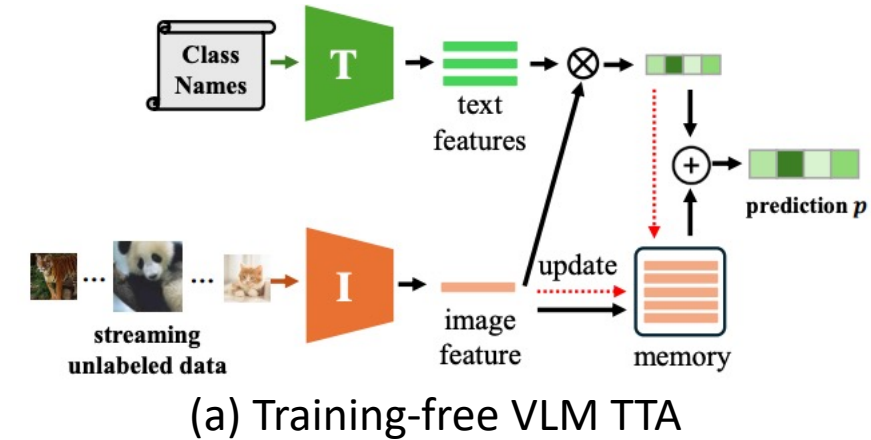


# VLM Test-time Adaptation: Can We Trust the Predictions?

- **VLM TTA:** Adapt pre-trained VLM to shifted test data without re-accessing source data and labels.
  - **Existing Methods: reliance on zero-shot predictions**
    - *Training-free:* confident zero-shot predictions as memory
    - *Training-based:* TTA loss on zero-shot predictions
  - **Question:** Can we trust zero-shot predictions? Will inaccurate predictions cause cascade failure in TTA?
- 

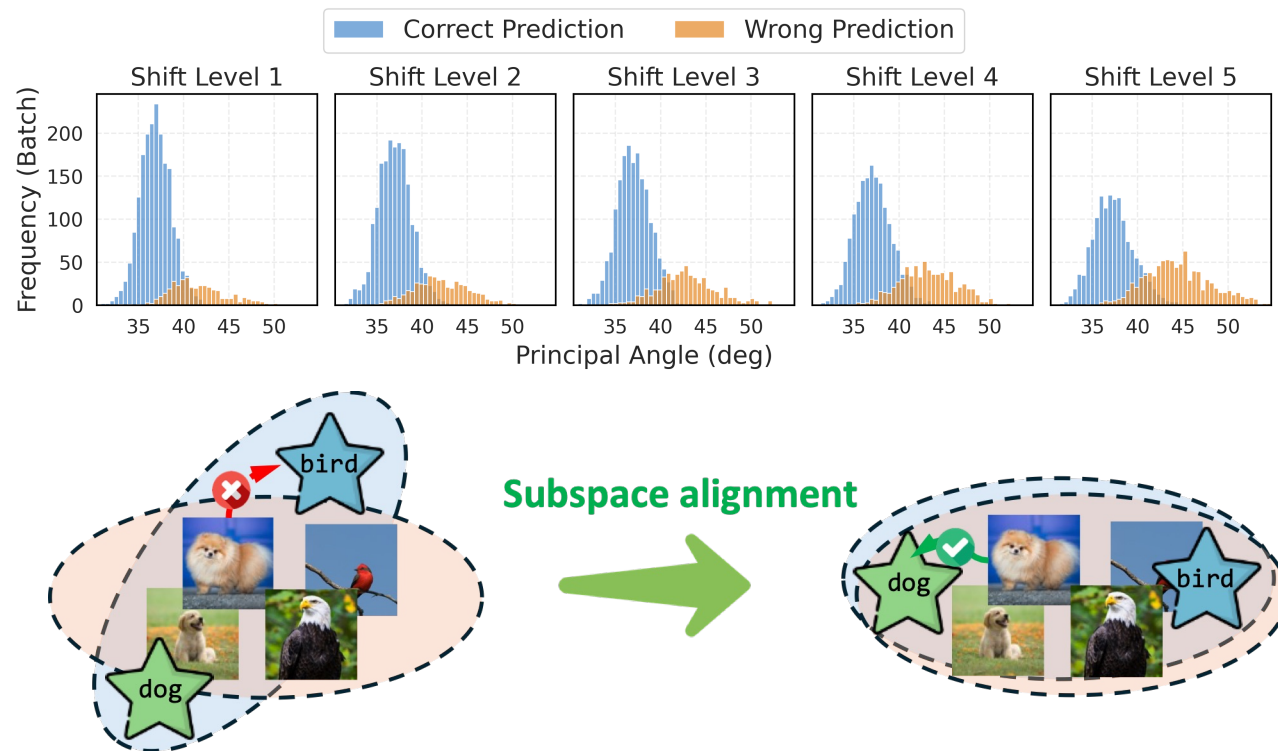
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graph LR
    A[Shifted images] --> B[Noisy zero-shot predictions]
    B --> C[Memory / self-training]
    C --> D[Reinforced errors]
  
```
- **Subspace Alignment For TTA (SubTTA):** Improve the cross-modal geometry before trusting the zero-shot prediction



# Observation 1: Modality Gap

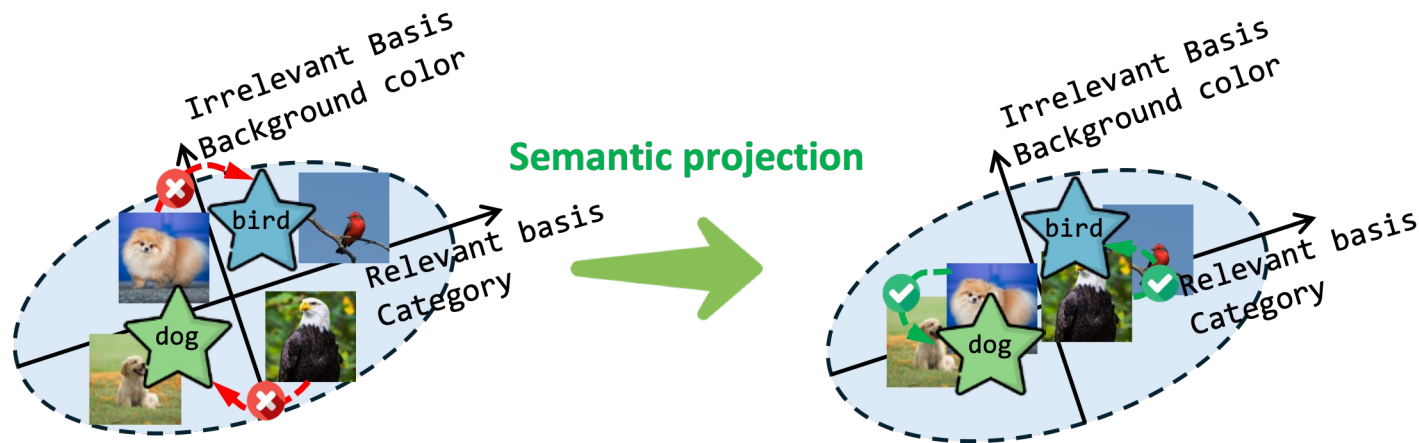
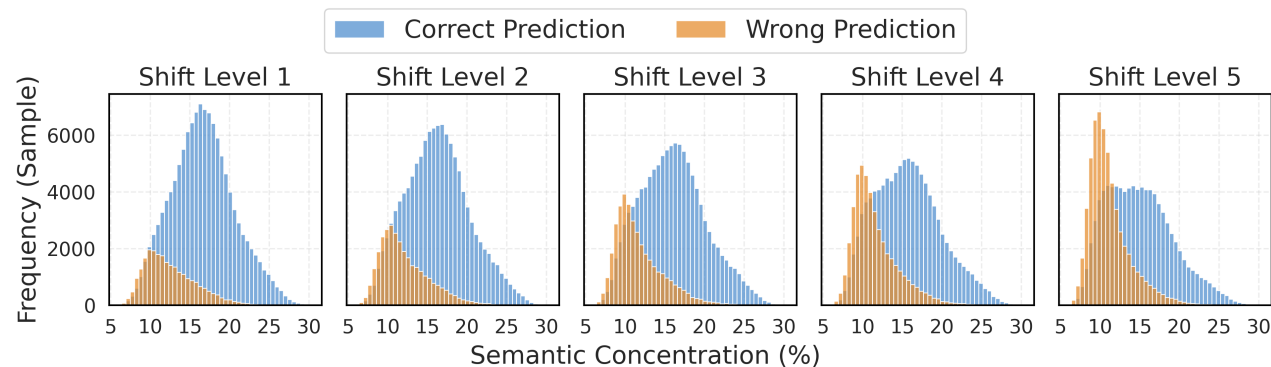
- **Metric:** principal angles between visual and textual subspaces ( $\downarrow$ ).
- **Observation:** Shifts exacerbate the gap between visual and textual modalities.
- **Example:** cross-modal shifts  $\rightarrow$  a dog image drifts toward the bird anchor  $\rightarrow$  incorrect zero-shot prediction.



**Solution: realign the visual subspace with the textual semantic anchor.**

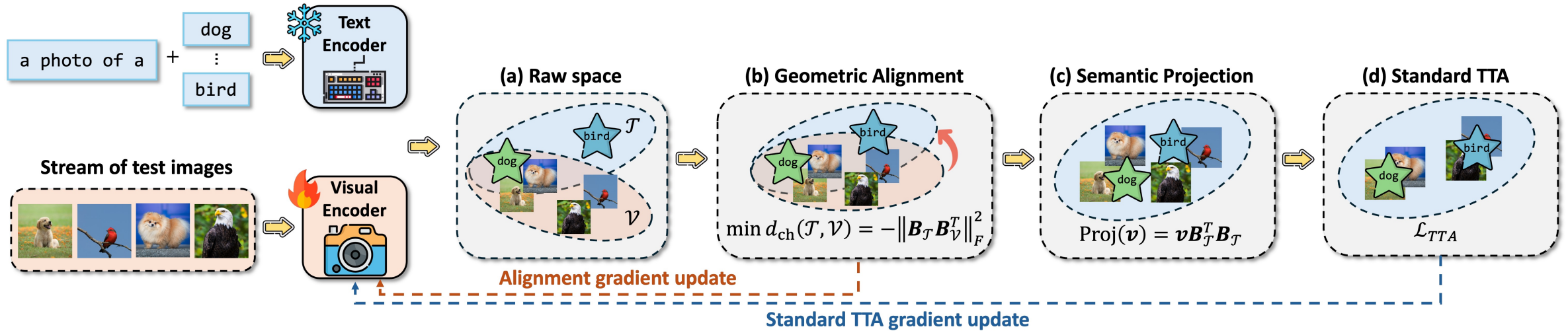
# Observation 2: Visual Nuisance

- **Metric:** semantic concentration—the fraction of visual energy within the textual subspace ( $\uparrow$ ).
- **Observation:** Shifts degrade semantic concentration of visual embeddings and exacerbate visual nuisances.
- **Example:** the blue sky creates a spurious bird cue, so a dog image is misclassified as a bird.



**Solution: suppress visual components outside the task-specific textual span.**

# Methodology: Align $\rightarrow$ Project $\rightarrow$ Adapt



## 1 Geometric Alignment

Bridge the modality gap.

## 2 Semantic Projection

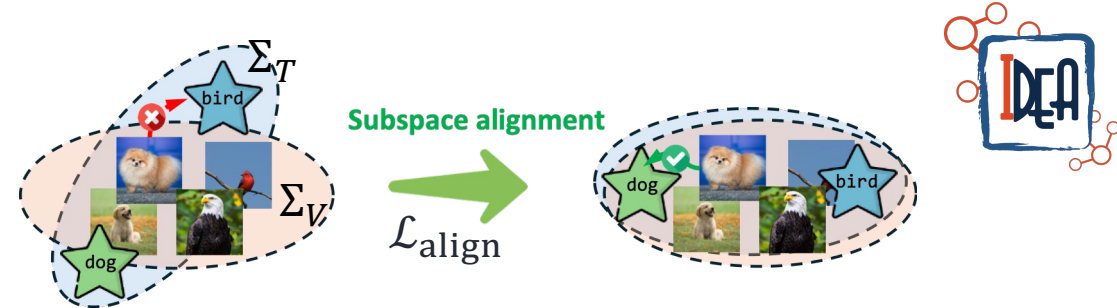
Filter visual nuisance.

## 3 Standard TTA

Refine decision boundaries.

**Objective: refine cross-modal geometry to improve zero-shot predictions for TTA.**

# Methodology: Subspace Alignment



## ➤ Subspace extraction

Covariance captures dominant feature directions.

$$\Sigma_T = T^\top T, \Sigma_{batch} = V^\top V$$

➤ **T / V**: normalized text / image features.

## ➤ Moving average

Mitigate small-batch noises/class incomplete by moving average mixing

$$\Sigma_V \leftarrow (1 - \alpha)\Sigma_V + \alpha\Sigma_{batch}$$

## ➤ Subspace bases

$$\Sigma_T = B_T^\top \Lambda_T B_T, \Sigma_{batch} = B_V^\top \Lambda_V B_V$$

## ➤ Geometric alignment

Min chordal distance → max cross-modal overlap

$$\mathcal{L}_{align} = r - \|B_T B_V^\top\|_F^2$$

- **Semantic anchor**: keep the textual basis fixed.
- **Update**: adapt visual normalization layers.
- **Basis invariance**: avoid rigid rank-to-rank matching.

# Methodology: Semantic Projection

## ➤ Semantic projection as a hard filter

$$v_{\text{proj}} = v B_T^\top B_T$$

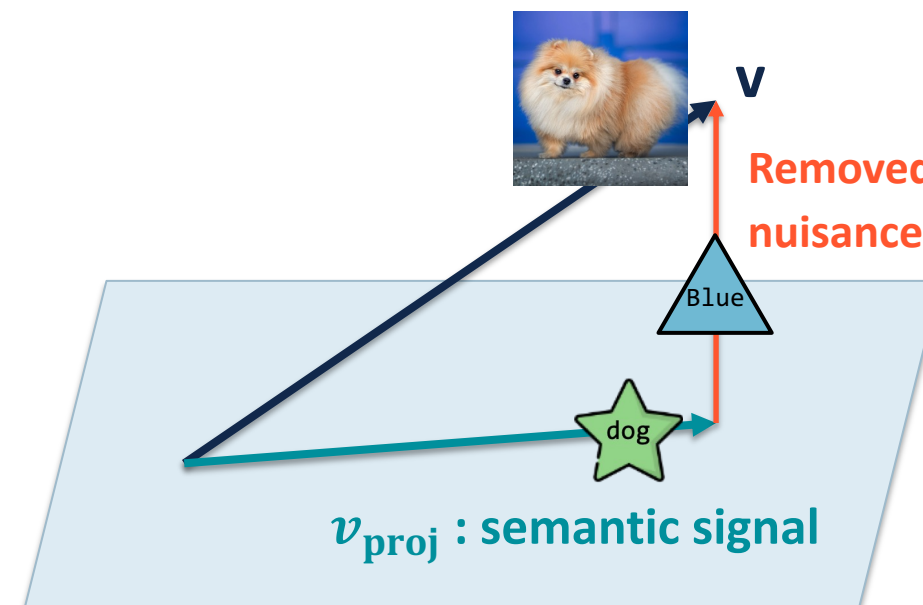
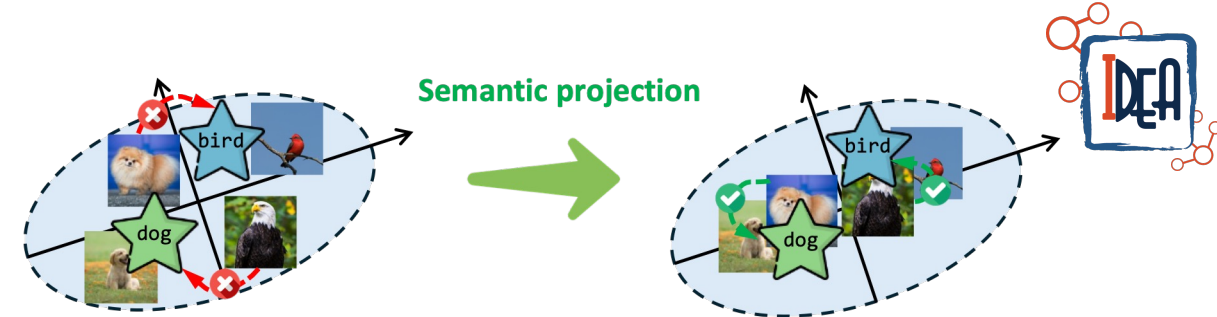
- **Retain:** task semantics within the textual span.
  - **Filter:** nuisance orthogonal to the span.
- **Adapt:** standard TTA on projected features.  
Subsequent test-time adaptation based on refined zero-shot predictions

**Entropy-based**

TENT

**Cluster-based**

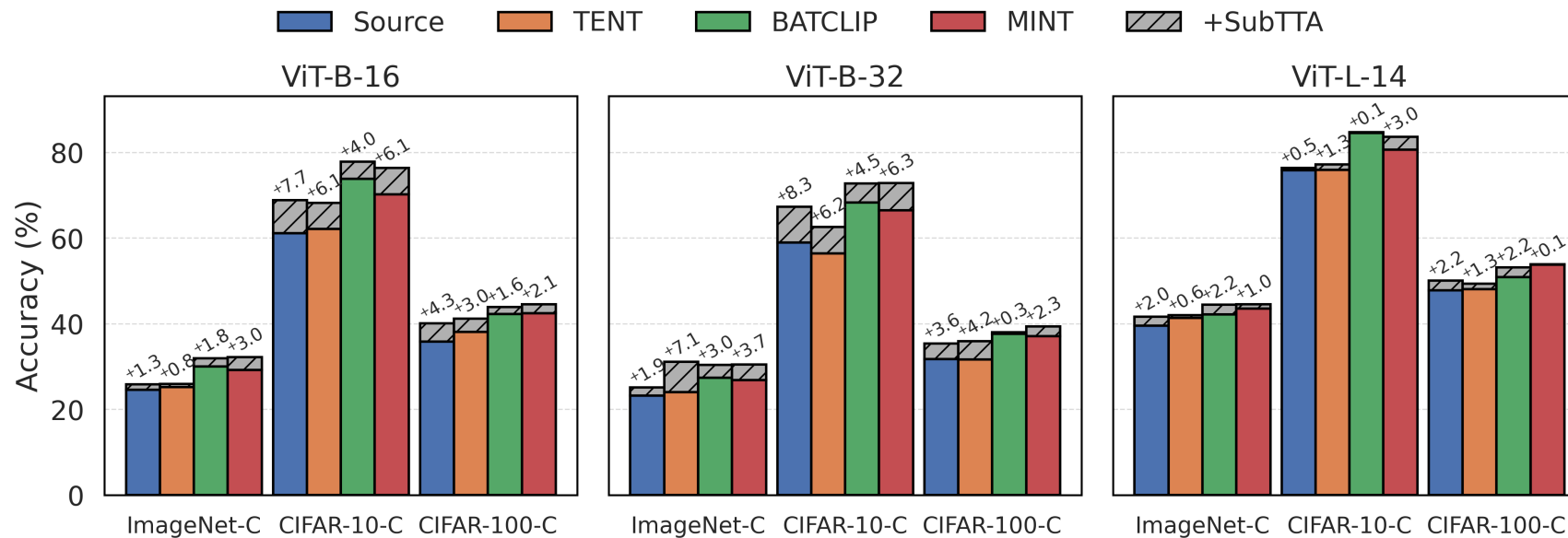
MINT / BATCLIP



**Task-specific textual subspace**

# Experiment: Benchmark Results

- **Experiment setting:** CIFAR/ImageNet; CLIP ViT-B-16 / B-32 / L-14; Image classification accuracy
- **Performance:** +2.24 pp on average; gains up to +8.3 pp.
- **Compatibility:** improves various TTA methods across CLIP backbones.
- **Robustness:** mitigates negative adaptation under different corruptions.



Zeng, Zhichen, et al. "Subspace alignment for vision-language model test-time adaptation." EMNLP 2026.

Wang, Dequan, et al. "Tent: Fully test-time adaptation by entropy minimization." ICLR 2021.

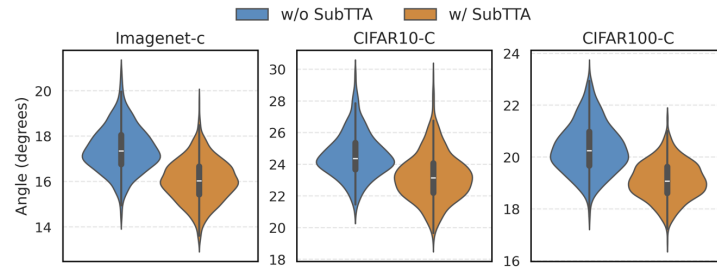
Bao, Wenxuan, Ruxi Deng, and Jingrui He. "Mint: A simple test-time adaptation of vision-language models against common corruptions." NeurIPS 2026.

Maharana, Sarthak, et al. "Batclip: Bimodal online test-time adaptation for clip." ICCV 2025.

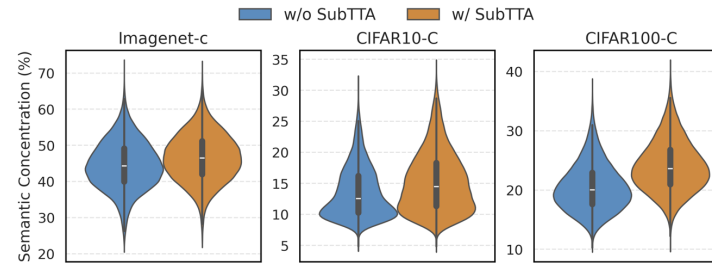
Radford, Alec, et al. "Learning transferable visual models from natural language supervision." ICML 2021.

# Experiments: Improved Geometry and Embedding

## Principal angles ↓

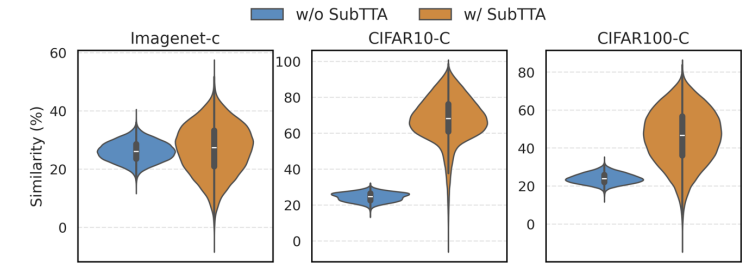


## Semantic concentration ↑



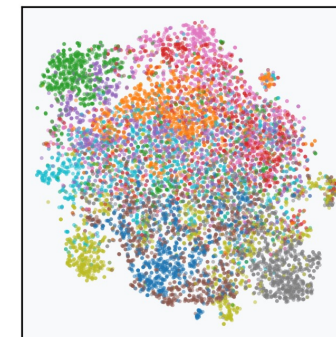
Blue: Source CLIP Orange: +SubTTA

## True-label similarity ↑

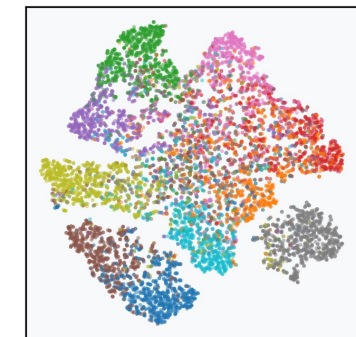


## Embedding quality

- **Compactness:** tighter within-class clusters.
- **Separation:** clearer boundaries between classes.



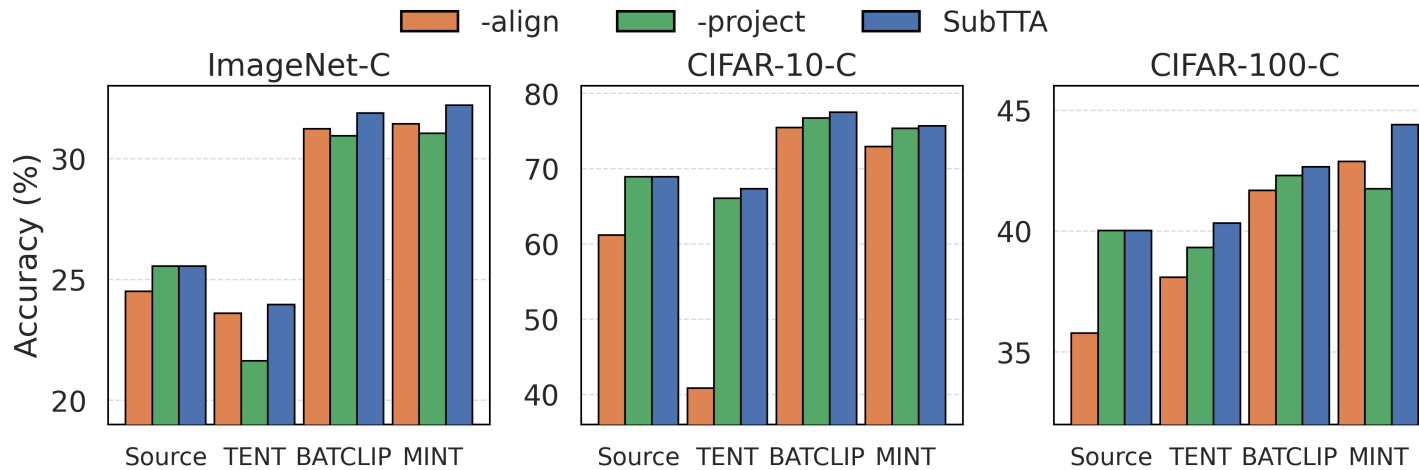
w/o SubTTA



w/ SubTTA

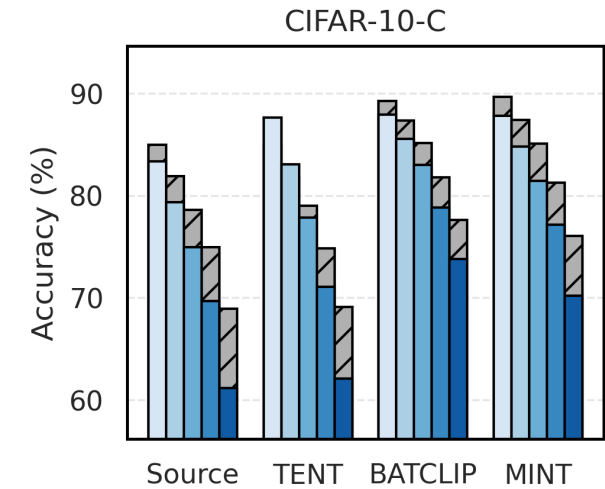
# Studies: Component Contributions and Shift Severity

## Ablation: alignment + projection



- **Without alignment:** projection may amplify prediction errors.
- **Without projection:** visual nuisance remains.

## Robustness across severity



Benefits persist across severity 1–5, with larger gains under stronger shifts.

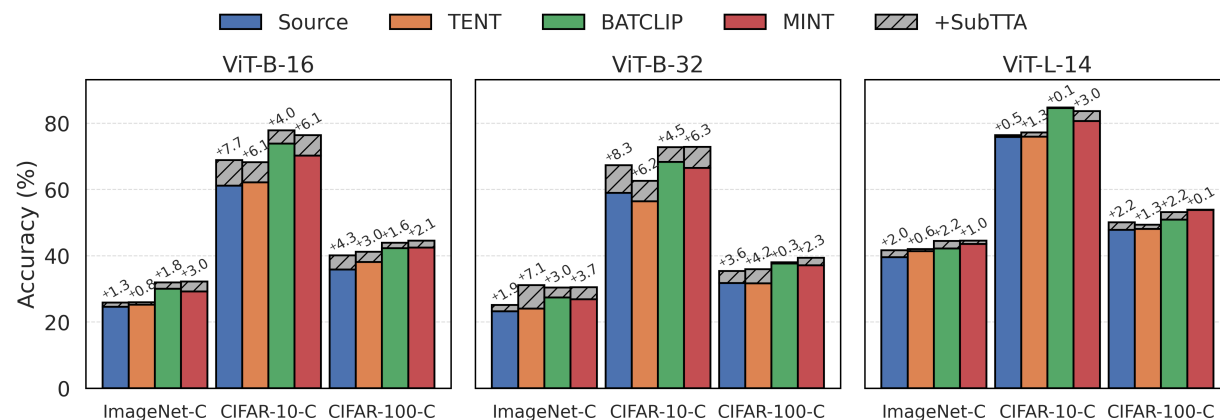
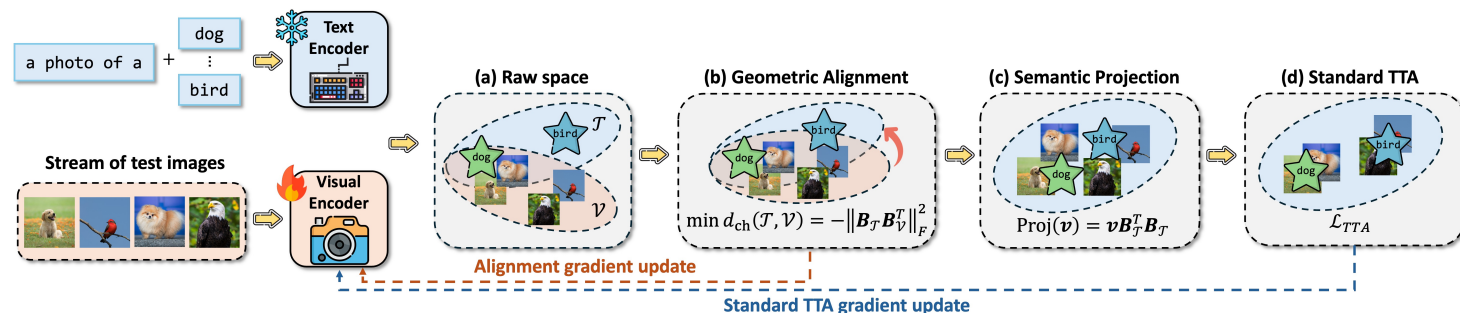
Colored: baseline; hatched: +SubTTA.

**Alignment and projection are complementary; benefits persist across shift severities.**

# Summary



- **Problem:** VLM test-time adaptation
- **Challenges:** unreliable zero-shot predictions
  - Modality gap
  - Visual nuisance
- **Key ideas**
  - Subspace alignment: mitigate modality gap
  - Semantic projection: suppress nuisance.
- **Results:** stronger performance and robustness
  - +2.24 pp on average across TTA methods.



Paper



Code



Slides



Poster

